Amdahl's Law:
Used for Computing Speedup of Parallel Processors

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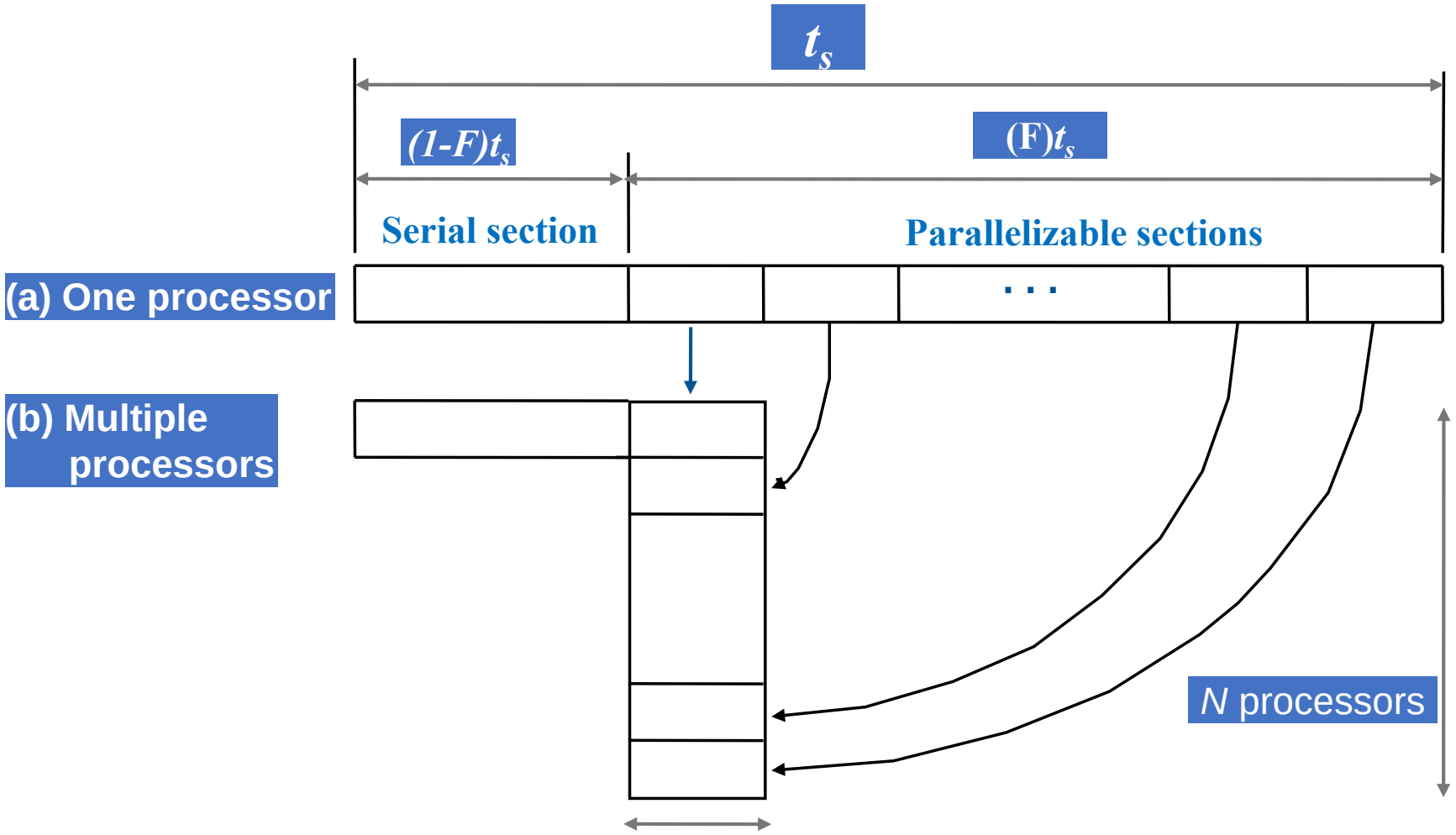
Amdahl's Law

- Amdahl's Law shows the limit of the speedup that can be derived by parallelizing an application.
- For an application, let F be the parallel fraction and (1-F) be the serial fraction.
- We use N processors to parallelize the parallel fraction. What is the speedup?

Maximum Speedup Amdahl's law



Compute Speedup – Amdahl's Law

N is the speedup on parallel fraction.

$$ExTime_{before} = [(1-F) + F]\ t_s$$

$$ExTime_{after} = [(1-F) + (F/N)]\ t_s$$

$$OverallSpeedup = \frac{ExTime_{before}}{ExTime_{after}} = \frac{1}{[(1-F) + \frac{F}{N}]}$$

Implications of Amdahl's Law

- Even with infinite number of processors, maximum speedup is limited to 1/(1-F).
- To attain high speedup, having high F (parallel fraction) is much more important than having high number of processors.
- If your application is not highly parallel, no point in trying to run it on a supercomputer and expect good performance.

Solved Question

- Compute speedup from parallelism for below values of F and N.

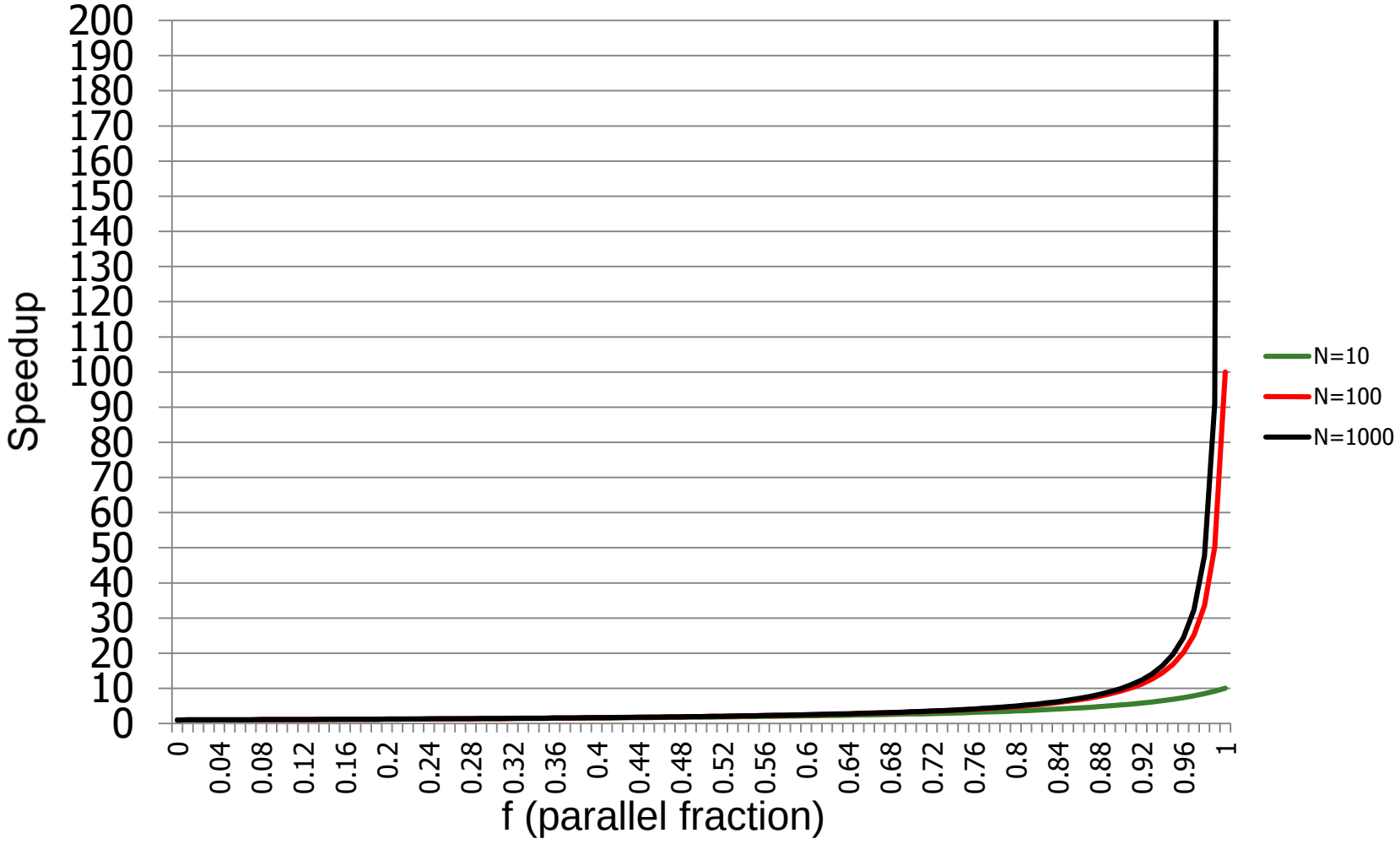
F	N	Speedup
0.2	10	
0.2	1000	
0.9	10	
0.9	1000	

Solved Question

- Compute speedup from parallelism for below values of F and N.

F	N	Speedup
0.2	10	1.22
0.2	1000	1.25
0.9	10	5.26
0.9	1000	9.91

Sequential Bottleneck



Solved Question

- If 89% fraction of an application is parallelizable, what is the maximum speedup one can get from parallelizing it.

- Solution: Let originally, the time be 100 seconds.
- Assuming infinite processor, the 89% of time can be reduced to zero, so only 11 seconds remain. The maximum speedup is 100/11 = 9.09

Solved question [1 of 2]

- An application spends 45% of time in initialization, 28% of time in vision-processing function and remaining time in signal-processing function. In system0, all the tasks are run on a single-core CPU.
- System1 has an signal-processing accelerator and a vision-processing accelerator which give a speedup of 8X and 5X, respectively over the single-core CPU execution. Find the speedup of system1 over system0 assuming that both the accelerators are used on system1.

Solved question [2 of 2]

	Time on System0	Speedup on system1	Time on system1
Initialization	45		45
Vision-processing	28	5X	5.6
Signal-processing	27	8X	3.375
Total	100		53.97

- Speedup = 100/53.97 = 1.85